

Facilitator Guide

TAKE NOTE

This Manual should be used in conjunction with the Learner Guide and the Learner Workbook

Unit Standard Title: Use mathematics to investigate and monitor the financial aspects of personal, business and national issues

Unit Standard No: 7456

Unit Standard Credits: 5

NQF Level: 3

Time allocation

Training: The following breakdown can be used for the program.

Session: 1 = 30 min	Study unit	Day	Type
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
	Summative assessment	Day___	

Time allocation

Block program: The block program is only a guideline to indicate what will be facilitated . It is important to note that the block program can be changed depending on the pace of the learners and how many learners attended the learning program.

DAY	08:00 TO 08:30	08:40 TO 09:10	09:20 TO 10:00		10:30 TO 11:00	11:10 TO 11:40	11:50 TO 12:30		13:00 TO 13:30	13:40 TO 14:10	14:20 TO 15:00		15:15 TO 16:00
1				T E A				L U N C H				T E A	
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
14													

Learning event plan

4

Use the format in table beneath to structure the content for a learning event. This provides guidelines for drawing up a facilitation plan in preparation for a presentation. An example of a facilitation plan is indicated in the next table.

Introduction	
Topic	♦ Select a suitable topic. ♦ Introduce the topic verbally. ♦ Display the topic visually.
Interest stimulus	♦ The aim of the interest stimulus is to obtain the learners' interest in the topic under discussion. ♦
Outcomes	♦ The outcome(s) state what the learners should be able to do after the learning event. ♦ Display the outcomes visually. ♦
Scope	♦ The scope of the presentation consists of all the main headings which will be covered in the theory lesson. ♦ Display the scope visually. ♦ Present the scope in a logical sequence. ♦
Importance	♦ The importance of the theory lesson must be stressed so that the learners realize how they will benefit by attending it. ♦
Ground rules	♦ The aim of establishing ground rules is to allow the learners to produce their own set of rules. ♦ Allow the learners to commit themselves to the ground rules by doing an observable activity. ♦
Physical environment	♦ The physical environment must be effective for learning. It includes aspects such as seating arrangement and equipment. ♦

Contents (Body)

Scope point no 1

The first scope point of the presentation will now be discussed in detail. Cover the rest of the scope points one by one.



- Establish a positive relationship with the learners (e.g. professional behavior).
- Indicate throughout the presentation which scope point is being discussed.
- Use at **least two relevant learning aids** (e.g. video, posters).
- The learning aids must support the theory.
- Present the theory lesson on the level of the target group.
- Explain the theory component by making use of appropriate examples.
- Allow the learners to ask questions or to comment.
- Apply effective verbal communication skills (e.g. speak loud enough, vary tone of voice).
- Apply effective non verbal communication skills (e.g. avoid irritating gestures, keep eye contact).
- Apply effective micro-skills (e.g. active listening, paraphrasing, summarizing).
- Control breathing so that voice usage is not affected by shortness of breath.
- Obtain maximum learner participation by including activities and group discussions.
- Keep control over disruptive influences (e.g. learners are not allowed to make unsuitable remarks and comments).
- Allow time for learners to ask questions during the presentation.

Closure

Summary

Questions

Assess

Interest stimulus

Summary

- ♦ Ask questions that are relevant to the outcomes stated.
 - ♦ Use different types of questions (e.g. close-ended, open-ended).
 - ♦ Avoid the use of questions such as leading, subjective, double-barreled etc.
 - ♦ Direct the questions to all the learners, instead of individuals.
 - ♦ Reinforce the questions and answers of the learners.
 - ♦ Provide positive feedback in time.
 - ♦ Allow activities to be performed to enable learners to reflect on the content.
 - ♦ Apply the theory to the practical situation (e.g. giving an example).
-
- ♦ Assess the learners by asking well selected questions concerning all the important aspects.



(It is very important that these questions are formulated in your planning phase and written into your presentation plan. It is thus determined whether the learners have achieved the outcome(s) of the theory lesson.)

- ♦ If there is a forthcoming presentation an interest stimulus must also be given to stimulate the interest of the learners for the next theory lesson.

Learning event plan

7

Study unit: _____				
Learning Event: _____				
Topic: _____				
Series No	Facilitator activities	Learner activities	Learning aids	Time
	INTRODUCTION			

Learning event plan

8

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Interest stimulus:			
	Outcomes: ♦ ♦ ♦ ♦ ♦			
	Scope:			
	Importance			

Learning event plan

9

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Ground Rules			
	Scope point no 1: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 2: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 3: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 4: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 5: (Contents)			

Learning event plan

14

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Summary:			

Learning event plan

15

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Assess:			
	Interest stimulus:			

Program evaluation questionnaire

16

Name of Facilitator: _____

Date: _____

Please give your honest reaction and comments. Mark the appropriate answer with an X

		Disagree	Neutral	Agree
LEARNING PROGRAM				
1.	The learning material is neatly compiled.			
2.	I received all the learning material in time.			
3.	The module content addressed my learning needs.			
4.	The exercises in the module enabled me to understand the theory.			
5.	The self assessments in the module enabled me to reach the stated outcomes.			
6.	The written assignment enabled me to master the content.			
7.	I could apply the theory into my work environment.			
LEARNING ENVIRONMENT				
8.	The venues were adequately prepared.			
9.	Suitable learning aids were used.			
LEARNING FACILITATORS				
10.	The learning facilitators demonstrated courteous behavior during training.			
11.	The learning facilitators were supportive towards me.			
12.	The learning facilitators appeared knowledgeable on the subject.			
13.	The learning facilitators demonstrated effective facilitation skills.			
14.	Enough time was allocated for me to practice facilitation skills.			
15.	I received adequate feedback on the progress that I made.			

Program evaluation questionnaire

17

PROGRAM CONTENT										
		1	2	3	4	5	6	7	8	9
16.	How do you rate the subject content?									
17.	How do you find the quality of this program to be?									
18.	Rate the program for the qualities listed:									
	• Practical value									
	• Thoroughness									
	• New ideas gained									
	• Maintain your interest									
	• Helpful to developing yourself									
	• Clear, understandable.									
	• Did you understand the outcomes of the program?									

GENERAL COMMENTS BY LEARNER:

ACTION TAKEN BY FACILITATOR:

LEARNER GUIDE

Unit Standard Title:	Use mathematics to investigate and monitor the financial aspects of personal, business and national issues
Unit Standard No:	7456
Unit Standard Credits:	5
NQF Level:	3

In terms of the Copyright Act 1978:

No part of this manual may be reproduced or translated in any form or by any means, electronic or mechanical including photocopying, recording or by an information storage and retrieval system without the express permission from TDA.

COPYRIGHT: TRAINING DYNAMICS AFRICA

First edition 2008

Publications disclaimer

Neither the authors nor the SETA can accept any responsibility for any legal or any other consequences, which may arise directly or indirectly as a result of the use or adaptation of or reliance upon any of the content of this material.

This material has been compiled based on the **UNIT STANDARD 7456: Use mathematics to investigate and monitor the financial aspects of personal, business and national issues**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

PROGRAM GUIDE	5-10
1. Introduction	5
2. Purpose of this learning program	5
3. Standards and qualifications	5
4. Assessments	6
5. Range statements	6
6. The learning guide	6
7. Learning assumed to be in place	7
8. Preparation of summative assessment	10
9. Roles and responsibilities	10
Study unit 1: Budgets	11-21
1. Budgets	13
2. Simple Interest (SI)	16
3. Compound Interest (CI)	17
4. Inflation	18
5. Appreciation and Depreciation	19
Glossary of Terms (Educational Terms)	23-24
Bibliography	25
SAQA Unit Standard	27-29

1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

This unit standard will be useful to people who aim to achieve recognition at some level in Further Education and Training or to meet the Fundamental requirement of a wide range of qualifications registered on the National Qualifications Framework.

People credited with this unit standard are able to:

- ♦ Use mathematics to plan and control personal and/or household budgets and income and expenditure.
- ♦ Use simple and compound interest to make sense of and define a variety of situations including investments, stokvels, inflation, appreciation and depreciation.
- ♦ Investigate various aspects of financial transactions including costs, prices, revenue, cost price, selling price, loss and profit.

2.1 Target group

This program is compiled for the following **target group**:

- ♦ Security members
- ♦ South African Defense Force members
- ♦ South African Police Force members
- ♦ Correctional Services
- ♦ Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

REMEMBER.....

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

REMEMBER.....

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

REMEMBER.....

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix
UNIT STANDARDS:

	ID	UNIT STANDARD TITLE	LEVEL	CREDITS
Core	244176	Use security equipment	Level 2	2
Core	114941	Apply knowledge of HIV/AIDS to a specific business sector and a workplace	Level 3	4
Core	244184	Apply legal aspects in a security environment	Level 3	8
Core	244177	Conduct a security patrol in an area of responsibility	Level 3	7
Core	117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No 60 of 2000) applicable to possessing a firearm	Level 3	3
Core	246694	Explain the requirements for becoming a security service provider	Level 3	4
Core	244182	Give evidence in court	Level 3	4
Core	244179	Handle complaints and problems	Level 3	6
Core	244181	Perform hand over and take over responsibilities	Level 3	2
Core	244189	Conduct access and egress control	Level 4	7
Core	242825	Conduct evacuations and emergency drills	Level 4	4
Core	11505	Identify, handle and defuse security related conflict	Level 4	12
Fundamental	119472	Accommodate audience and context needs in oral/signed communication	Level 3	5
Fundamental	9010	Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations	Level 3	2
Fundamental	9013	Describe, apply, analyse and calculate shape and motion in 2- and 3-dimensional space in different contexts	Level 3	4
Fundamental	119457	Interpret and use information from texts	Level 3	5
Fundamental	9012	Investigate life and work related problems using data and probabilities	Level 3	5
Fundamental	119467	Use language and communication in occupational learning programmes	Level 3	5
Fundamental	7456	Use mathematics to investigate and monitor the financial aspects of personal, business and national issues	Level 3	5
Fundamental	119465	Write/present/sign texts for a range of communicative contexts	Level 3	5
Elective	117889	Demonstrate an understanding of Victim Empowerment	Level 2	6
Elective	12484	Perform basic fire fighting	Level 2	4
Elective	13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5
Elective	244178	Apply neighbourhood watch principles	Level 3	5
Elective	113852	Apply occupational health, safety and environmental principles	Level 3	10
Elective	243188	Care for a service dog	Level 3	15
Elective	116534	Carry out basic first aid treatment in the workplace	Level 3	2

UNIT STANDARD 7456: Use mathematics to investigate and monitor the financial aspects of personal, business and national issues



Elective	113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5
Elective	114942	Describe how to manage reactions arising from a traumatic event	Level 3	2
Elective	243190	Handle a trained service dog to deter crime	Level 3	8
Elective	119649	Handle and use a handgun	Level 3	2
Elective	119651	Handle and use a manually operated rifle or carbine	Level 3	2
Elective	119650	Handle and use a self-loading rifle or carbine	Level 3	2
Elective	119652	Handle and use a shotgun	Level 3	2
Elective	114979	Operate a computer workstation in a business environment	Level 3	2
Elective	244335	Conduct security at an event	Level 4	5
Elective	252174	Demonstrate an understanding of security measures within a port security environment	Level 4	4
Elective	246693	Explain the requirements for safeguarding a national key point	Level 4	4
Elective	123515	Handle and use a handgun for business purposes	Level 4	3
Elective	123519	Handle and use a manually operated rifle or carbine for business purposes	Level 4	3
Elective	123511	Handle and use a self loading rifle or carbine for business	Level 4	3
Elective	123514	Handle and use a shotgun for business purposes	Level 4	3
Elective	11985	Inspect, drive and maintain an official vehicle	Level 4	3
Elective	11513	Operate effectively within a specified control room environment	Level 4	15

8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

Budgets

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Use mathematics to plan and control personal, regional and/or national budgets and income and expend.
- ♦ Use simple and compound interest to make sense of and define a variety of situations.
- ♦ Use mathematics to debate aspects of the national economy.

TAKE NOTE

The secret of being financially strong is not necessarily to earn more money, but to *use the money you have* in a wiser way. This will involve learning to draw up a **budget**.

1. Budgets

DEFINITION

A budget is a *plan of how money will be spent*. It should clearly show the balance between income (money in) and expenditure (money out).

The object of a budget is to make sure that you:

- ♦ Spend less than you earn
- ♦ Keep as much money as possible for yourself
- ♦ Save some money for your future in the form of investments

1.1 Personal or family budget

In a personal or family budget all sources of income (inflows) are identified and expenses (outflows) are planned with the intent of matching outflows to inflows (making ends meet). It is a finance plan that allocates future personal income towards expenses, savings and debt repayment - it allocates or distributes expected income to expected expenses and intended savings.

- ♦ **Expense categories:** List your expenses under three categories:
 - ⇒ **Fixed expenses:** These are the regular items, such as your home loan repayments or rent, vehicle or loan repayments, insurance and school fees
 - ⇒ **Variable expenses:** These are regular expenses the amount of which varies, for example, the amount you spend on electricity and water, food, petrol or transport, doctor and dentist bills, gifts and entertainment
 - ⇒ **Irregular expenses:** These are usually annual amounts, such as car and television licences and school uniforms. Don't forget to budget for additional tax if you regularly have to pay in when your tax is assessed. If you don't know how to work out the tax you are likely to pay, ask a financial or tax adviser to help you.

- ◆ **Essential budgeting rules:** Here are some other fundamental rules for correct and effective budgeting. These are:

- ⇒ **Keep your budget realistic:** Part of the budgeting exercise is to help you identify what you are spending on things. A budget helps you to identify where you are spending too much, and to rein in your spending habits.

However, a budget will not help you control your finances if it is not realistic. If, for example, you have tried to limit your grocery shopping to the essentials and cut out all the luxury grocery items, yet your spending still exceeds what you have budgeted for groceries, then you may have to set aside more money for groceries and cut out an expense somewhere else in your budget.

If you exaggerate any part of your budget, it should be rather to underestimate your income (if it varies from month to month) and to overestimate your spending. If you do the opposite, you will just be setting yourself up to fail.

- ⇒ **Get your priorities right:** Distinguish between your needs and your wants. Budget for your needs first and your wants last. Set your expenses in order from most to least important to you. Cut out the least important expense items first. In deciding your priorities, one of your first and most important actions should be to repay your debts because debts cost you more money in the form of the interest you have to repay.

Pay off the debt that is incurring the highest interest first - for example, the debt you owe on your credit card. Once you have cleared shorter-term debt, try to budget for extra repayments for longer-term debt, such as your home loan.

Another priority should be for you to save. You must regard saving as one of your expenses, but the advantage of saving is that it is spending on yourself and your future financial security.

However, do not put an inordinate amount of money into one area at the cost of another as this will only make budgeting more difficult.

- ⇒ **Budget for a surplus:** Do not budget to spend every cent you have. If you can, leave a margin of about five percent of your income to cushion yourself against your expenses amounting to more than you expected.

A good idea is to budget for luxuries, such as a night out, using the surplus money you saved from the previous month. This is also a way of rewarding yourself for having got your budget right.

- ⇒ **Involve your family:** If you have a family, let the members of your family who are old enough take part in your budgeting exercise.

By taking part in the process, your family members will come to appreciate why some expenses need to be cut and may even demand less money from you. It will also help them to learn to budget with their own money one day.

- ⇒ **Create an emergency fund:** You must expect that you will not always get your budget right and that unforeseen events will incur unforeseen expenses. You should try to set some savings aside to protect you in the event of an emergency, such as when your car breaks down, your house is burgled, or a member of your family is sick and the treatment is not covered by health insurance, or if you lose your job.

An emergency is not the impulsive purchase of luxuries that are not included on your budget.

- ◆ **Revisit your budget:** Once you have drawn up your budget and are spending in accordance with it, you need to revisit it on a monthly basis.

Do not get disheartened if it is not working out straight away because it can take up to 18 months to make a budget work.



You must save and not just blow whatever money you earn each month!

1.2 Business/Corporate budget

◆ Business Start-up Budget

The process of calculating the costs of starting a small business begins with a list of all necessary purchases including tangible assets (for example, equipment, inventory) and services (for example, remodeling, insurance), working capital, sources and collateral. The budget should contain a narrative explaining how you decided on the amount of this reserve and a description of the expected financial results of your business activities.

◆ Corporate budget

The budget of a company is compiled annually. A finished budget usually requires considerable effort and can be seen as a financial plan for the new financial year. While traditionally the Finance department compiles the company's budget, modern software allows hundreds or even thousands of people in the various departments (operations, human resources, IT etc) to contribute their expected revenues and expenses to the final budget. If the actual numbers delivered through the financial year turn out to be close to the budget, this will demonstrate that the company understands their business and has been successfully driving it in the direction they had planned. On the other hand, if the actuals diverge wildly from the budget, this sends out an 'out of control' signal and the share price could suffer as a result.

1.3 National budget

The government needs people and businesses to pay part of their income or profits for the use of all people in South Africa. These payments, called taxation, form part of the income of the government. The government can also borrow money from the International Monetary Fund (IMF), the World Bank, or other countries, but the interest rates are high and it must be able to repay the loans over time. The government uses this income (the taxes) to provide services for the benefit of all the people of South Africa. The expenditure required for these services is huge, so it is the duty of all people to pay the taxes they owe.

2. Simple Interest (SI)

DEFINITION

Simple interest is the amount of interest paid on the capital (investment) amount only, at the end of every year.

EXAMPLE:

Michael invests R2000.00 at an interest rate of 10% p.a. (per annum/annually) simple interest for 4 years.

- ◆ After 1 year, Michael's investment will have grown as follows:

$$10\% \text{ of R2000.00} = \text{R200.00}$$

- ◆ After 4 years, he would have earned:

$$\text{R200.00} \times 4 = \text{R800.00} \text{ interest on his investment}$$

- ◆ His investment amounts to:

$$\text{R2000.00} + \text{R800.00} = \text{R2800.00} \text{ in total after 4 years.}$$

We can use a formula to calculate this:

$$SI = \frac{P \times r \times t}{100}$$

Where: **P = Principal** (the amount of money invested or borrowed)

r = interest rate (given as a percentage i.e. $\frac{\text{---}}{100}$)

t = time (in years) for which the principal has been invested

So Michael earns:

$$SI = \frac{2000 \times 10 \times 4}{100}$$

=R800.00 interest

His investment amounts to: R2000.00 + R800.00 = R2800.00 after 4 years

3. Compound Interest (CI)



Compound interest is when the interest earned on the money invested (principal) is added at the end of each time period to the principal, and interest is then earned on the accumulated amount. It is interest added to interest previously earned on a principal amount. That means, the interest is not taken out and spent, so it **compounds or accumulates all the time**.

There is a formula which is used for compound interest:

$$A = P\left(1 + \frac{r}{100}\right)^n$$

Where: **A = Amount at the end of the investment period**

P = Principal (invested at the beginning of the investment period)

r = rate (in percentage per annum)

n = number of periods (not necessarily years)

EXAMPLE:

Let's use the formula to calculate Michael's interest:

- ♦ $P = R2000$
- ♦ $r = 10$
- ♦ $n = 4$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$= 2000\left(1 + \frac{10}{100}\right)^4$$

$$= 2000(1,1)^4$$

$$= 2928.2$$

$$\approx R2928.20$$

- ♦ His **investment** amounts to: **R2928.20**
- ♦ His **interest** earned is: $R2928.20 - R2000.00 = R\ 928.20$

4. Inflation

DEFINITION

In mainstream economics, inflation means a rise in the general level of prices of goods and services over time. It is therefore, a large, continuous increase of prices in the economy, but buyers don't get anything extra for the money they spend.

The result is that people use more money on consumables and less on savings. Inflation is normally shown as a percentage. For example, we might hear that inflation this year is 10%. What this means is, if a basket of food cost you R100 last year, this year it will cost you 10% more. I.e. the same basket of food will cost R110 this year ($R100 + 10\% = R110$)

EXAMPLE:

- ♦ Susan works as a cashier in a supermarket. She tries to save the same amount of money every month. However, inflation gets in the way:
 - ⇒ **In 2007:** Susan earned R100.00 a day and she usually worked 25 days per month. So her gross monthly income was: **$25 \times R100.00 = R2\ 500.00$**
 - ♦ Susan was careful with her money and managed to keep her expenses to R2 000.00 per month, so she saved R500.00 every month.
 - ⇒ **In 2008:** Susan's monthly salary was increased by 5%.
5% of R2 500.00 = $0,5 \times R2\ 500.00 = R125.00$
So Susan's monthly salary for 2008 was:
 $R2\ 500.00 + R125.00 = R2\ 625.00$
 - ♦ In the same year, the inflation rate was estimated to be about 8%. As a result of inflation, Susan's expenses increased by 8%.
 - ⇒ 8% of R2 000.00 = $0,8 \times R2\ 000.00 = R160.00$
So Susan's monthly expenses for 2008 were:
 $R2\ 000.00 + R160.00 = R2\ 160.00$
 - ⇒ If we subtract her expenses from her income in 2008:
 $R2\ 625.00 - R2\ 160.00 = R465.00$
- It is clear that Susan saves less every month in 2008, even though she had a salary increase and her expenses stayed the same. It is all as a result of inflation!

5. Appreciation and Depreciation

DEFINITION

- ♦ **asset** - anything owned by a person/organisation that has monetary value
- ♦ **appreciate** - increase in value
- ♦ **depreciate** - decrease in value
- ♦ **book value** - what is noted and accepted as the value of a vehicle

Some assets appreciate, whereas others, although they may be valuable, might depreciate.

EXAMPLE:

A house is an asset. Usually it will appreciate in value and you will be able to sell it for more than the price you purchased it at.

Cars are also assets. However, they begin to depreciate the moment you drive them out of the showroom. After ten years, the book value of a car is no longer noted, as the depreciation is so great that the car is almost worthless. There are exceptions like certain vintage or exotic cars.

We can calculate how much an asset has appreciated in the same way that we work out compound interest on investment, i.e. by using compound interest calculations.

The formula $A = P(1 + \frac{r}{100})^n$ is used for the effects of *inflation* and for *appreciation* (since it is also compounded).

If we need to calculate the value of something which has *depreciated*, we use the following version of the formula: $A = P(1 - \frac{r}{100})^n$

EXAMPLE:

A car costs R98 300 and depreciates by 8% annually, calculated on the diminishing balance. What will the car be worth after 5 years?

$$\begin{aligned} A &= R98\,300(1 - \frac{8}{100})^5 \\ &= R98\,300(1 - 0,08)^5 \\ &= R98\,300(0,92)^5 \\ &= R98\,300(0,6590814) \\ &= \mathbf{R64\,787.70} \end{aligned}$$

[illegible]

WORDS/ACRONYMS	EXPLANATION OF ACRONYMS
ASSESSMENT	“A structured process of gathering evidence and making judgments about an individual’s performance in relation to registered national standards and qualifications”
ASSESSMENT CRITERIA	It describe the evidence that is needed to show that the learner has demonstrated the outcome correctly. A list of evidence to be supplied by the candidate as proof of his / her ability to integrate all of the competencies listed earlier in the standard.
COMPETENCE	Competence is the ability to do certain things in a range of different environments with knowledge and understanding about what is being done and the effect this has on others. It includes <u>Knowledge, skills, attitudes, values and ethics.</u>
CREDIT	Each unit standard is worth a number of credits which gives an indication of the size and scope of the unit standard, the standard is worth. (1 credit = 10 notional hours of learning)
CRITICAL CROSSFIELD OUTCOMES	critical cross field outcomes is to direct educational activities towards development of the learners within a social and economic environment.
EMBEDDED KNOWLEDGE	Body of theory essential for understanding and applying the standard. Principles, methods and techniques that would underpin good practice of this unit standard.
EVIDENCE	Evidence is a set of information or observations that provides proof of the learner’s competence.
FORMATIVE ASSESSMENT	Aims to assist the learner to identify any knowledge/skills gaps when demonstrating his/her current competence. Diagnostic assessment supports learning and progress of the learner.
INTEGRATED ASSESSMENT	Integrated assessment refers to: • assessing a number of outcomes together • assessing a number of assessment criteria together • assessing a number of unit standards together • using a combination of assessment methods and instruments for an outcome/s • collecting naturally occurring evidence in the workplace • Acquiring evidence from other sources such as supervisor’s reports, testimonials, portfolio of work previously done, logbooks, journals etc.
LEARNING ASSUMED TO BE IN PLACE	Building blocks (knowledge, skill and understanding) assumed to be in place
LEARNING PROGRAM	A “learning program is a set of learning activities in which the learner will be involved in working towards the achievement of one or more outcomes.
LEVEL	Each unit standard is registered on the NQF at a particular level based on the complexity of the outcomes and stringency of the standards.
NQF	National Qualifications Framework
RANGE STATEMENTS	The range statements indicate detailed requirements of the assessment criteria.

RE-ASSESSMENT	• The same method and instrument may be used, but, the task and materials should be different • The task and materials should, however, be of the same complexity and level as the previous ones. • In case the methods and instruments are changed it must be ensured that they are appropriate for the outcomes specified.
RPL ASSESSMENT	RPL is a way of recognizing what individuals already know and can do. The basic premise underlying it is that people, especially mature adults, learn many things inside and outside the formal structures of education and training and that irrespective of where, how and when learning was acquired, subject to assessment, it can be worthy of recognition and credit.
RPL	Recognition of prior learning
SASSETA	Safety & security sector education and training authority.
SAQA	SOUTH AFRICAN QUALIFICATION AUTHORITY
SPECIFIC OUTCOMES	An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention. Key performance areas in the standard are given, the focus is <i>not</i> only on current competence, but also on future competence in this standard.
SUMMATIVE ASSESSMENT	Assessment used to record the level of competence achieved by a learner over a period of time, e.g. at the end of a program, term or year.
QUALIFICATIONS	Represent a planned combination of learning outcomes which has a defined purpose and intended to provide learners with applied competence and a basis for further learning
UNIT STANDARDS	the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF). “A statement of the outcomes that the learner must achieve to earn credit. Standards are nationally agreed and internationally comparable.”

Use mathematics to investigate and monitor the financial aspects of personal, business and national issues

SAQA US ID		UNIT STANDARD TITLE	
7456		Use mathematics to investigate and monitor the financial aspects of personal, business and national issues	
ORIGINATOR		REGISTERING PROVIDER	
SGB Math Literacy, Math, Math Sciences L 2 -4			
FIELD		SUBFIELD	
Field 10 - Physical, Mathematical, Computer and Life Sciences		Mathematical Sciences	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular-Fundamental	Level 3	5
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2008-02-26	2011-02-26	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2012-02-26		2015-02-26	

This unit standard does not replace any other unit standard and is not replaced by any other unit standard.

PURPOSE OF THE UNIT STANDARD

This unit standard will be useful to people who aim to achieve recognition at some level in Further Education and

Training or to meet the Fundamental requirement of a wide range of qualifications registered on the National Qualifications Framework.

People credited with this unit standard are able to:

Use mathematics to plan and control personal and/or household budgets and income and expenditure.

Use simple and compound interest to make sense of and define a variety of situations including investments, stokvels, inflation, appreciation and depreciation.

Investigate various aspects of financial transactions including costs, prices, revenue, cost price, selling price, loss and profit.

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

The credit value is based on the assumption that people starting to learn towards this unit standard are competent in Mathematics and Communications at NQF level 2.

UNIT STANDARD OUTCOME HEADER

Use mathematics to plan and control personal, regi

Specific Outcomes and Assessment Criteria:

SPECIFIC OUTCOME 1

Use mathematics to plan and control personal, regional and/or national budgets and income and expend

OUTCOME NOTES

1.1 Plans describe projected income and expenditure realistically.

1.2 Calculations are carried out using computational tools efficiently and correctly and solutions obtained are verified in terms of the context.

1.3 Budgets are presented in a manner that makes for easy monitoring and control.

1.4 Actual income and expenditure is recorded accurately and in relation to planned income and expenditure. Variances are identified and explained and methods are provided for control.

OUTCOME RANGE

Bank accounts, provincial and key elements of national budgets and tax.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

Use mathematics to plan and control personal, regional and/or national budgets and income and expenditure.

ASSESSMENT CRITERION NOTES

1.1 Plans describe projected income and expenditure realistically.

1.2 Calculations are carried out using computational tools efficiently and correctly and solutions obtained are verified in terms of the context.

1.3 Budgets are presented in a manner that makes for easy monitoring and control.

1.4 Actual income and expenditure is recorded accurately and in relation to planned income and expenditure. Variances are identified and explained and methods are provided for control.

ASSESSMENT CRITERION RANGE

Bank accounts, provincial and key elements of national budgets and tax.

SPECIFIC OUTCOME 2

Use simple and compound interest to make sense of and define a variety of situations.

OUTCOME NOTES

2.1 The differences between simple and compound interest are described in terms of their common applications and effects.

2.2 Methods of calculation are appropriate to the problem types.

2.3 Computational tools are used efficiently and correctly and solutions obtained are verified in terms of the context or problem.

2.4 Solutions to calculations are used effectively to define the changes over a period of time.

OUTCOME RANGE

Effective and nominal rates, commission, appreciation and depreciation.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

Use simple and compound interest to make sense of and define a variety of situations.

ASSESSMENT CRITERION NOTES

2.1 The differences between simple and compound interest are described in terms of their common applications and effects.

2.2 Methods of calculation are appropriate to the problem types.

2.3 Computational tools are used efficiently and correctly and solutions obtained are verified in terms of the context or problem.

2.4 Solutions to calculations are used effectively to define the changes over a period of time.

ASSESSMENT CRITERION RANGE

Situations include:

Effective and nominal rates, commission, appreciation and depreciation.

SPECIFIC OUTCOME 3

Use mathematics to debate aspects of the national economy.

OUTCOME NOTES

3.1 Values are calculated correctly.

3.2 Mathematical tools and systems are used effectively to determine, compare and describe aspects of the national economy.

3.3 Debating points are based on well-reasoned arguments and are supported by mathematical information.

OUTCOME RANGE

Tax, productivity and the equitable distribution of resources.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

Use mathematics to debate aspects of the national economy.

ASSESSMENT CRITERION NOTES

3.1 Values are calculated correctly.

3.2 Mathematical tools and systems are used effectively to determine, compare and describe aspects of the national economy.

3.3 Debating points are based on well-reasoned arguments and are supported by mathematical information.

ASSESSMENT CRITERION RANGE

Tax, productivity and the equitable distribution of resources.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

The following essential embedded knowledge will be assessed by means of the specific outcomes in terms of the stipulated assessment criteria. Candidates are unlikely to achieve all the specific outcomes, to the standards described in the assessment criteria, without knowledge of the listed embedded knowledge. This means that the possession or lack of the knowledge can be inferred directly from the quality of the candidate's performance against the standards.

Budgets.

Terminology and definitions associated with financial situations.

Estimation and approximation.

Compound increase and decrease.